Week 4 Notes

Busy Beaver

Jacks discussion of Turning machines, and the discussion of beaver tripples, reminds me of the busy beaver problem. Which is to create the longest terminating program using minimal Turing machine states.

From Busy Beaver wikipedia.

> Deciding the running time or score of the nth busy beaver is noncomputable.

> An implication of the busy beaver game is that, if it were possible to compute the functions Σ(n) and S(n) for all n, then this would resolve all mathematical conjectures that can be reduced to a halting problem, e.g., into a form of "does ⟨this Turing machine⟩ halt".[6] For example, there is a 25-state Turing machine that checks Goldbach's conjecture for each number and halts on a counterexample; if this machine did not halt after running for S(25) steps, then it must run forever, resolving the conjecture.[7] Many other problems, including the Riemann hypothesis (120[8] states) and the consistency of ZF set theory (745 states[9][10]), can be expressed in a similar form, where at most a countably infinite number of cases need to be checked.[6]

I am however pretty sure that this is already well explored, and there are no low hanging fruit.

Mentor content

TODO: Find geometric representation for multi party computation.

TODO: Try to find bounds for the voting problem.

TODO: look into Busy Beaver

TODO: look into other models of computation

TODO: watch more lectures